

ACTIVITY 7.

Identification of power cables, electrical wiring components and instruments:

-Cables and wires

What are cables?

The standard definition of a cable is a bundle of two or more insulated wires that are protected by an outer sheath or jacket. Cables are designed to transmit electrical power, signals, or data from one location to another.

Some common types of cables include:

- 1. Power cables** - used to transmit electrical power from one location to another, typically in high-voltage applications such as power distribution networks.
- 2. Coaxial cables** - used to transmit high-frequency signals such as those used in cable TV and internet connections.
- 3. Ethernet cables** - used to transmit data between devices on a local network, such as a home or office network.
- 4. USB cables** - used to connect electronic devices such as smart-phones, computers, and cameras to each other or to a power source.
- 5. HDMI cables** - used to transmit high-definition video and audio signals between devices such as TVs, DVD players, and gaming consoles.

What are wires?

The standard definition of a wire is a single, flexible, cylindrical strand or rod of metal, typically made of copper or aluminum, that is used to transmit electrical power or signals from one location to another.

Wires come in many different types and sizes, and they are often categorized based on their gauge (thickness), insulation type, and intended use. Some common types of wires include:

- 1. Solid wires** - have a single, solid core and are typically used in electrical applications where a reliable, low-resistance connection is needed.
- 2. Stranded wires** - made up of multiple thin wires twisted together, and are typically used in applications where flexibility is required, such as in cables or wiring harnesses.
- 3. Coaxial wires** - consist of a central conductor surrounded by a dielectric insulator and an outer conductor, and are commonly used for high-frequency signal transmission.
- 4. Ribbon wires** - made up of multiple conductors arranged side-by-side, and are typically used for low-voltage applications such as internal wiring in electronic devices.

Differences between a wire and a cable.

Although the terms "wire" and "cable" are often used interchangeably, there are some key differences between the two:

- 1. Construction:** A wire is typically a single conductor made of a conductive material such as copper or aluminum, while a cable is a bundle of two or more wires enclosed in an outer sheath or jacket.
- 2. Functionality:** A wire is generally used to carry a single electrical signal or power, while a cable can carry multiple signals or power sources.
- 3. Insulation:** A wire may or may not be insulated, depending on the application. In contrast, cables are almost always insulated to protect the wires from damage and to prevent electrical interference.
- 4. Size:** Wires are generally smaller in diameter than cables, which can contain multiple wires of varying sizes.
- 5. Application:** Wires are typically used for short-distance, low-voltage applications such as within electronic devices or in residential wiring, while cables are commonly used for long-distance, high-voltage applications such as power transmission or in commercial or industrial settings.

Where are wires used?

- In electronic appliances
- In electrical wiring
- In electrical power distribution over short distances

Where are cables used?

- Power transmission over long distances
- Telecommunications
- Networking

Power Cables:

- A power cable is an assembly of two or more electrical conductors held with an overall sheath.
- Its general use is the transmission and distribution of power.
- It is installed in buildings, buried underground or run overhead.
- Flexible ones used for portable devices, mobile tools and machinery.

Components of Power Cable

- ◆ **Core (conductor)** : it transports current. The core of a cable can be made up of 1,2,3 or 4 conductors, which are typically made of aluminium or copper.
- ◆ **Insulation** : it covers the conductors and prevent them from coming into contact with each other, or with other materials. The insulation also protects the conductors from external environmental factors such as moisture, heat and abrasion. For low voltage cables, the insulation materials is typically made of PVC (polyvinyl chloride) or similar materials. For medium and high voltage cables, the insulation material is made up of materials such as XLPE (cross-linked polyethylene), EPR (ethylene propylene rubber), or TR-XLPE (tree-retardant cross linked polyethylene).

- ◆ **Sheath** : it encases the core and the insulation. The main function of the sheath is to protect the cable from mechanical damage, moisture, chemicals and other environmental factors.
- ◆ **Bedding** : the bedding protects the cable against corrosion and it provides a smooth surface for the sheath to be applied to. It also fills any gap or voids between the insulation and the sheath
- ◆ **Armour** : it provides rigidity and protects the cable from mechanical injury. It also provides some degree of electrical shielding or grounding.
- ◆ **Serving** : it is a layer of material applied over the armour. Its function is to provide additional protection over environmental and atmospheric conditions such as moisture, heat and abrasion.

Classification of power cables on the following basis:

Number of cores:

- 1. Single core cable:** this type of cable has only one core conductor. It is commonly used for high-voltage transmission and distribution.
- 2. Multi-core cable:** this type of cable has multiple cores or conductors. It is commonly used for low-voltage power distribution, as well as data transmission and communication applications.
- 3. 3 core cable**
- 4. 4 core cable**

Voltage and frequency for which they are manufactured:

- 1. Low-voltage cables** - designed for use in low voltage applications (up to 1000V). they are commonly used for power distribution and lighting in residential and commercial buildings.
- 2. Medium voltage cables:** these cables are designed for use in medium voltage applications (from 1000V to 35,000V). they are

commonly used for power transmission and distribution in industrial and utility settings.

3. High voltage cables: these cables are designed for use in high voltage applications (above 35,000V). they are commonly used for long distance power transmission and distribution, as well as for high voltage industrial applications.

Types of insulating materials:

1. PVC insulated cables: for low voltage applications in residential and commercial buildings.

2. XLPE (Cross-linked Polyethylene): for medium and high voltage applications in industrial and utility settings.

3. Insulated rubber cables: commonly used in portable and temporary applications, such as for construction sites or events venues.

4. EPR (Ethylene Propylene Rubber) insulated cables: commonly used for medium and high voltage applications in industrial and utility settings.

5. Fibreglass insulated cables

Type of protection:

1. Unarmored cables: have no armor and rely on their insulation to protect the conductors. They are used in indoor applications.

2. Armored cables: have armor. Armored cables are used in outdoor and industrial applications where there is a risk of physical damage.

Electrical wiring components:

1. Wires

2. Fuses

3. Switches

4. Lamps

5. Circuit breakers

6. Junction and outlet boxes

7. Resistors, capacitors and inductors:

8. Integrated circuits

9. Conduit: this is a tube or channel used to protect wires and cables from physical damage and exposure to environmental factors.

Electrical wiring instruments:

1. Cable tie tools: used to tighten and cut cables, also known as zip ties or wire ties.

2. Electrical testing tools: used to measure and test electrical systems for performance, functionality, and safety.

3. Clamp tools: used to secure and connect electrical cables and wires in place.

4. Wire strippers: used to remove the insulation from electrical wires so that they can be connected to other wires.

5. Fuse pullers: used to remove fuses from an electrical panel or other fuse holder. They are designed to safely remove a fuse without damaging the fuse or the surrounding components.

6. Fish tape: this is a flexible tool used to pull wires through conduit, walls, and ceilings.

7. Cable ties: these are plastic ties used to organize and secure electric cables and wires.